Technology Stack and Rationale

Trip Planner — Supporting Project Document

React

React provides a component-based way to build interactive forms, result panels and saved-trip views.

TypeScript

TypeScript helps describe the shape of trip inputs and API responses and catches some data-shape errors during development.

Vite

Vite provides the frontend development server and build tooling. The production build should be checked before sharing a release.

Python and FastAPI

Python hosts the backend integration logic. FastAPI exposes HTTP endpoints and supports request validation and structured responses.

Groq

Groq is used as the model API provider for generating travel-planning text. Availability and response quality depend on provider access, model behavior and quotas.

Supabase and PostgreSQL

Supabase provides managed database access backed by PostgreSQL. Saved-trip persistence depends on schema compatibility and appropriate server-side permissions.

Markdown Rendering

Markdown rendering makes headings and lists in generated content easier to read. Model output should still be checked for malformed or misleading content.

Configuration

Secrets belong in backend environment configuration and must not be committed. Dependency files should describe what is required to run the project.